

# Grey Matter

## FitRank — Explainable AI Candidate Ranking

### PROBLEM STATEMENT

Recruiters searching 100,000 anonymized candidate profiles need a top-100 shortlist that is accurate, adversarial-resistant, and explainable — not a keyword-stuffed list that surfaces honeypot profiles (e.g., “HR Manager” with 9 ML skills) above genuine Senior AI Engineers.

### TEAM

**Shubh Varshney**

Team Leader

**Vedant Gupta**

Team Member

**Offline · Explainable · CPU-only · <5 min on 100K profiles**

# Solution Overview

## What is your proposed solution?

- Offline, two-stage ranking system: scores all 100K candidates vs a JD, returns top 100 with a human-readable reasoning string each.
- Stage 1 — Heuristic features: 28 interpretable signals (JD fit, career ML depth, coherence, skill trust, Redrob signals, penalties).
- Stage 2 — Learned re-ranking: LightGBM on pseudo-labels (top 5% positive, bottom 20% negative, traps forced negative).
- Output: submission.csv (candidate\_id, rank, score, reasoning) + audit\_report.json.

## What differentiates your approach?

- JD-conditioned, not JD-blind — parses JD into a RoleProfile before scoring.
- Profile coherence engine — catches title×skill mismatches & honeypot patterns.
- Semantic capability matching (all-MiniLM-L6-v2), not keyword counting.
- Redrob-native signals + explainable by default + adversarial-resistant (12 penalties, honeypot ×0.25).

### Traditional keyword matching vs FitRank

| Traditional                   | FitRank                                |
|-------------------------------|----------------------------------------|
| Counts AI keywords in summary | Career evidence + semantic embeddings  |
| Ignores title–skill coherence | Coherence engine + 12 penalty types    |
| No platform verification      | Redrob assessments, GitHub, engagement |
| No explanation                | Structured reasoning per candidate     |

# JD Understanding & Candidate Evaluation

## Key requirements extracted from the JD

*Target: Senior AI Engineer — Founding Team @ Redrob AI (Series A)*

- Production embeddings retrieval (sentence-transformers, BGE, E5)
- Vector DB / hybrid search (Pinecone, FAISS, Weaviate, Milvus)
- Ranking & recommendation systems at product companies
- LLM fine-tuning (LoRA, QLoRA, PEFT) — nice-to-have
- 5–9 yrs with production ML depth, not tutorials
- Pune/Noida preferred; Tier-1 India; ≤30-day notice
- Disqualifiers: pure research, consulting-only, CV/speech w/o NLP, title-chasers, skill inflation

## Most important candidate signals

- 1 Career ML depth
- 2 Current-role evidence (RAG, embeddings, FAISS)
- 3 Profile coherence (title domain × skills)
- 4 Skill trust (tenure + endorsements)
- 5 Redrob assessments (LoRA, PEFT, RAG)
- 6 Platform trust (GitHub, completeness)
- 7 Availability (OTW, notice, relocate)
- 8 Penalty flags (inflation, honeypot)

## Fit beyond keyword matching

- Title gate → ML\_AI | AI\_ADJACENT | SOFTWARE | NON\_TECH
- Dense embeddings compare full candidate text vs JD
- Career analyzer scores ML depth with per-role caps & momentum (tanh)
- Coherence engine checks templates, title-domain & skill-career consistency
- JD-adaptive penalties — disabled when JD domain matches

# Ranking Methodology

## Retrieve → Score → Rank

- Load 100K candidates from JSONL (streaming, schema-validated)
- Parse JD → RoleProfile (titles, capabilities, seniority, disqualifiers)
- Extract 28 RankingFeatures via parallel heuristic modules
- Score: heuristic formula OR LightGBM learned ranker (auto)
- Min-heap of top 100 —  $O(N \log 100)$
- Deterministic sort + reasoning + write CSV/JSON + validate

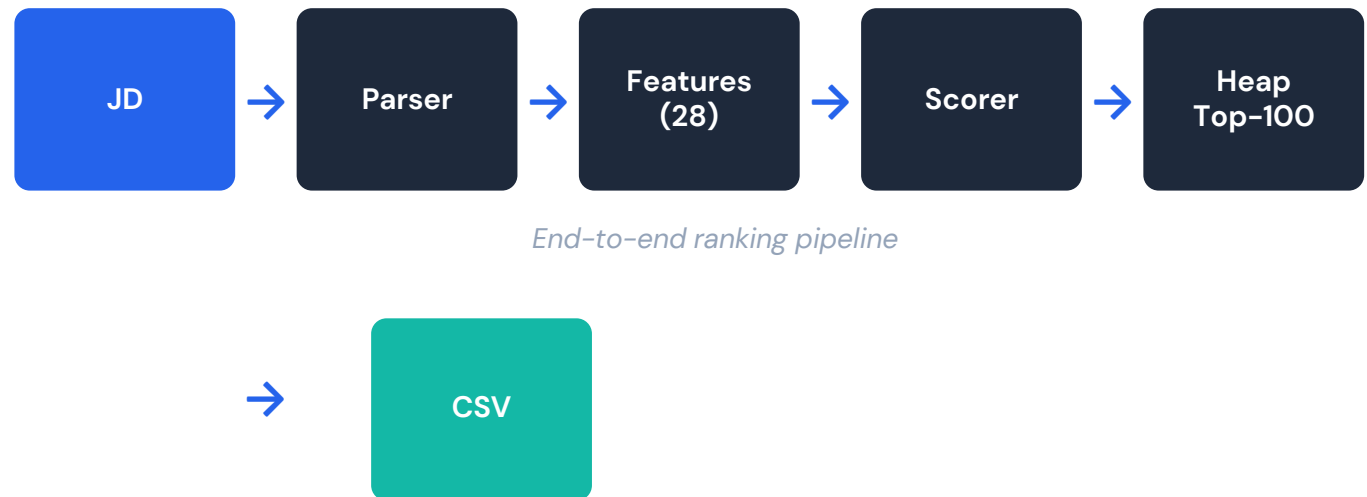
## Models, algorithms & heuristics

- JD Parser, Title Gate, Career Analyzer, Coherence Engine
- Skill Trust chain + Semantic Embedder (MiniLM 384-dim)
- Penalty Layer (12 traps) + LightGBM binary ranker
- Optional: local Qwen2-1.5B GGUF for NL reasoning (top-100 only)

## Combining signals — heuristic formula

$$\text{Final} = 0.25 \times \text{JD} + 0.30 \times \text{Career} + 0.20 \times \text{Coherence} + 0.15 \times \text{Trust} + 0.10 \times \text{Avail} - \text{Penalties}$$

Penalties capped  $0.70 \cdot \text{honeypot} \times 0.25$  if penalty  $> 0.30$  · Learned mode blends 55% LightGBM + 45% heuristic for stable ordering.



# Explainability & Data Validation

## How ranking decisions are explained

Rank #1 · CAND\_0071974 · score 0.95

Senior AI Engineer | JD=0.85 Career=0.58 Coh=0.79 Trust=0.85 | current role: embedding,Pinecone,recommendation; embedding,Pinecone,recommendation; assessments LoRA=87, PEFT=86; GH 83; 7.8yrs total/7.7yrs ML India (OTW/45d notice)

Format: {title} | JD=X.XX Career=X.XX Coh=X.XX Trust=X.XX | {evidence} — 100/100 strings unique & candidate-specific.

## Preventing hallucinations

- Default reasoning is template-based — built ONLY from extracted features, never invented credentials.
- Optional local LLM uses facts-only prompts + regex guard (must include JD=X.XX), falls back to template if invalid.
- No cloud API calls during ranking — external\_api\_calls\_during\_ranking: false.

## Handling suspicious profiles — 12 penalty types + honeypot detection

- Template mismatch · skill inflation · shallow boilerplate · expert-zero-endorse
- Consulting-only · pure research · wrong domain (CV/speech) · salary inverted
- Experience gap · job hopper · seniority mismatch · domain mismatch

### ✓ GENUINE #1

CAND\_0071974

Senior AI Engineer · score 0.95

Production RAG + embeddings, LoRA=87, 7.7 yrs ML.  
Included in top 100.

### ⚠ HONEYPOT TRAP ×0.25

CAND\_0004989 · CAND\_0000339

Title-skill mismatch, salary inversion, template stuffing.

Excluded from top 100. Genuine probe  
CAND\_0033861 included.

# End-to-End Workflow



```
$ python rank.py --candidates data/candidates.jsonl --jd data/job_description.txt --out outputs/submission.csv
```

# System Architecture

CLI: rank.py

Demo: streamlit\_app.py

**RANKER** `fitrank/ranker.py` — Heap Top-100 | Heuristic / LightGBM / Auto

**FEATURE EXTRACTOR** `fitrank/features.py` — 28 interpretable RankingFeatures / candidate

JD Parser /  
Title Gate

Career  
Analyzer

Coherence  
Engine

Penalties  
(12 types)

**EMBEDDER** — all-MiniLM-L6-v2

**SIGNALS** — Redrob platform

**DATA** `candidates.jsonl · job_description` **CONFIG** `weights.yaml · normalization.yaml`

OFFLINE: train LightGBM · precompute embeddings    OPTIONAL: local LLM reasoning (top-100 only)

## LAYERED MODULES

Each module is file-backed for traceability.

**No network during ranking** — all models cached locally.

# Results & Performance

<5 min

on 100K, CPU

0

honeypots in top 100

100/100

explained & unique

## Submission validation

- `validate_submission.py` → “Submission is valid.”
- 100 rows, scores monotonically decreasing (0.95 → 0.05)
- Known honeypots excluded: CAND\_0004989, CAND\_0000339
- Genuine probe included: CAND\_0033861 · 0 honeypot flags
- 19 unique job titles in top 100 (diverse, not single-title)

## Stability & data pool insights

- Top-20 overlap under ±10% perturbation: mean 19.7/20 · Jaccard 0.97 · 100% stable runs
- Pool: ~905 (0.9%) genuine ML · 4,337 keyword stuffers · 18,865 salary-inversion · 50,844 (~51%) title-career mismatch

## Top-5 snapshot

|    |              |                    |      |
|----|--------------|--------------------|------|
| #1 | CAND_0071974 | Senior AI Engineer | 0.95 |
| #2 | CAND_0055992 | AI Engineer        | 0.87 |
| #3 | CAND_0081846 | Lead AI Engineer   | 0.87 |
| #4 | CAND_0018499 | Senior ML Engineer | 0.86 |
| #5 | CAND_0079387 | AI Engineer        | 0.85 |

| Constraint    | FitRank Result                     |
|---------------|------------------------------------|
| CPU only      | ✓ sentence-transformers + LightGBM |
| No network    | ✓ all models pre-downloaded        |
| ≤5 min / 16GB | ✓ ~284s w/ pre-computed embeds     |
| Deterministic | ✓ heap + candidate_id tie-break    |
| Tests         | ✓ 223 fast + 6 slow (229)          |

# Technologies Used

## Core Language & Runtime

- Python 3.11+ — typed dataclass pipeline
- NumPy — feature vectors & embedding math

## Ranking & ML

- LightGBM — offline binary ranker, fast CPU
- sentence-transformers (MiniLM, 384-dim)

## Data & Config

- JSONL streaming loader (100K, low RAM)
- JSON Schema validation · YAML weights

## Semantic & Search

- Cosine similarity on dense embeddings
- Sparse capability fallback within budget

## Explainability

- Template reasoning engine (validator-safe)
- llama-cpp + Qwen2-1.5B GGUF (optional)

## Demo & Validation

- Streamlit + Plotly explorer + radar
- pytest (229) · validate\_submission.py

## Why these choices

**CPU-only stack** meets NFR (no GPU). **Offline pre-computation** makes ranking network-free. **Interpretable + learned blend** is explainable AND data-driven. **LightGBM over deep ranker** trains <1 min, no overfit on synthetic data.

# Submission Assets

**GitHub Repository** [github.com/brainRottedCoder/redrob-datachallenge](https://github.com/brainRottedCoder/redrob-datachallenge)

[make install](#) · [download-model](#) · [precompute](#) · [train](#) · [run](#) · [validate](#)

## Key Deliverables

- outputs/submission.csv — top-100 + reasoning
- outputs/audit\_report.json — runtime, top-5
- deck/Redrob\_FitRank\_Approach.pdf
- submission\_metadata.yaml · validate\_submission.py

## Interactive Demo

- streamlit run app/streamlit\_app.py
- JD paste, radar charts, penalty breakdown
- CSV export from shortlist
- Optional local-LLM AI reasoning toggle

## Documentation

- README.md — install, reproduce, constraints
- PRD\_Redrob\_FitRank.md — 13-phase design
- DELIVERABLES\_CHECKLIST.md — PRD §12

**Grey Matter | Shubh Varshney (Team Leader) | Vedant Gupta (Team Member)**

GREY MATTER

# Thank You

**FitRank** — Intelligent Candidate Discovery & Ranking

*"Trustable ranking that survives adversarial profiles."*

**Shubh Varshney** · Team Leader

**Vedant Gupta** · Team Member

[github.com/brainRottedCoder/redrob-datachallenge](https://github.com/brainRottedCoder/redrob-datachallenge)

**100K**

scored

**<5 min**

runtime

**0**

honeypots in top 100